Day 1: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

A1-1

For real x, explain why dividing both sides of an equation by x can lose a solution. Give one example.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A3-1

If r and s are the roots of $ax^2 + bx + c=0$, $a \neq 0$, what are r+s and rs? Explain by expanding.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I1-1

What happens to an inequality when both sides are multiplied by a negative number? Give a numerical example.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F1-1

What does $(f \circ g)(x)$ mean, and which inputs are allowed?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L1-1

If a statement says “P implies Q”, what are its converse and contrapositive? Which is logically equivalent to the original?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S1-1

State the nth term and sum of the first n terms of an arithmetic sequence with first term a and common difference d, for positive integer n.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C1-1

When should counts of possibilities be multiplied, and when should they be added? State the disjointness condition.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N3-1

List the repeating last digits of positive powers of 7. Which cycle position corresponds to an exponent divisible by 4?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____